Product Definition Document

DESCRIPTION OF REQUIREMENT(S)

Title:
Enhanced NEM12 (eNEM12) File Format – Including Hot Water

Background:
The eNEM12 file format requires further modification to accommodate interval data for Hot Water usage.

Purpose:
The purpose of this document is to define the format and structure for the additional information required in the eNEM12 file to support Hot Water interval data with a view to gaining agreement prior to undertaking IT development work.

Requirements:

1) File Format
   a) With the exception of the additional requirements specified in this document, the file format will comply with the MDFF specification as defined and amended from time-to-time by AEMO.
   b) The file will contain interval data (and associated aggregated records) for the following commodities only:-

Table 1

Commodity	Unit(s) of Measure	NMI Suffix(es)
Electricity	kWH	B*, E*
Hot Water	kL	W*

   c) Any 400 and 500 records will be omitted from the file.

2) Hot Water Data
   a) Where configured, the solution will include and present the Suffix and UOM in the eNEM12 file for a hot water data stream as follows:-
      i) Suffix = 'W1'; and
      ii) UOM = 'kL'.

      It is acknowledged that the W1 Suffix is being used contrary to the NMI Procedure.

3) Actual Data Only
   a) The solution will deliver 'Actual' data only. Any 'IntervalDate' for which estimated (E) and/or substituted data (S/F) exists will be excluded from the output file.|

4) Interval Length
   a) All data will be delivered in the 'as metered' interval length – The file may contain a mixture of 15 and/or 30 minute interval data.

5) NMI Allocation
   a) Where PLUS ES is appointed as the embedded network manager (MYENM), all customers will be allocated a NMI in preparation for publication to MSATS but will only be published to MSATS in the event the customer moves to on-market.

i) The solution will store Market NMI to non-Market NMI relationships similar to that detailed in the following table:-

Table 2	Market NMI	Non-Market NMI	From Date	To Date
	7105000001	ZZZZ000001	01/01/2018	31/12/9999
	7105000002	ZZZZ000002	01/01/2018	31/12/9999
	7105000003	ZZZZ000003	01/01/2018	31/12/9999

ii) The solution will check for the presence of the Market NMI in MSATS and if not present, or if present but NMI Status = ‘N’ for the ‘IntervalDate, present the non-Market NMI in lieu of the Market NMI in the eNEM12 file.

iii) The solution will interpret non-Market NMIs in accordance with the following table:-

Table 3	Characters	Length	Population Rules
	01-03	3	‘ZZZZ’
	04-07	4	Unique to Specific Embedded Network
	08-10	3	Unit Number (001-999)

b) Where PLUS ES is not appointed as the embedded network manager (not MYENM), the embedded network operator will provide PLUS ES with the non-Market NMI to Market NMI relationships (where applicable).

i) The embedded network operator will also be responsible for advising PLUS ES as NMIs move from on market to off market and vice versa.

6) <u>600 Record</u>

a) A 600 record will be included immediately following each 300 record where the corresponding 200 record contains a UOM = ‘kWH’ and a Suffix = ‘B*’ or ‘E*’.

b) The 600 record will contain the following information separated by a comma (,):-

Table 4	Field	Format	Population Rules
	Record ID	Numeric(3)	‘600’
	NMI	Char(10)	<u>NMI</u>* from 200 record
	Meter Serial Number	VarChar(12)	<u>MeterSerialNumber</u>* from 200 record
	Date	Date(8)	<u>IntervalDate</u>* from 300 record
	Channel ID	Char(2)	<u>NMISuffix</u>* from 200 record
	Daily Summation	Numeric(15.3)	Sum of interval values in 300 record
	Total Summation	Numeric(15.3)	Sum of previous day’s ‘Total Summation” and current day’s ‘Daily Summation’. Starting value to be zero (0).

** Indicates as specified in the NEM File Format Specification*

Example: 600,ZZZZ000001,214243202,20170302,E1,5.111,721.308

c) The 'Total Summation' value will be reset following a meter change and will be representative of the total consumption recorded on the current (or new) meter.

7) 610 Record

a) A (new) 610 record will be included immediately following each 300 record where the corresponding 200 record contains a UOM = 'kL' and a Suffix = 'W*'.

b) The (new) 610 record will contain the following information separated by a comma (,):-

Table 5	Field	Format	Population Rules
	Record ID	Numeric(3)	'610'
	NMI	Char(10)	NMI* from 200 record
	Meter Serial Number	VarChar(12)	MeterSerialNumber* from 200 record
	Date	Date(8)	IntervalDate* from 300 record
	Channel ID	Char(2)	NMISuffix* from 200 record
	Daily Summation	Numeric(15.3)	Sum of interval values in 300 record
	Total Summation	Numeric(15.3)	Sum of previous day's 'Total Summation" and current day's 'Daily Summation'. Starting value to be zero (0).

** Indicates as specified in the NEM File Format Specification*

Example: 610,ZZZZ000001,214243202,20170302,W1,5.111,721.308

c) The 'Total Summation' value will be reset following a meter change and will be representative of the total consumption recorded on the current (or new) meter.

8) 700 Record

a) A 700 record will be included immediately following each 600 record where the corresponding 200 record contains a UOM = 'kWH' and a Suffix = 'E*' (only).

b) The 700 record will contain the following information separated by a comma (,):-

Table 6	Field	Format	Population Rules
	Record ID	Numeric(3)	'700'
	NMI	Char(10)	NMI* from 200 record
	Meter Serial Number	VarChar(12)	MeterSerialNumber* from 200 record
	Date	Date(8)	IntervalDate* from 300 record
	Channel ID	Char(2)	NMISuffix* from 200 record
	Daily Summation	Numeric(15.3)	'1'
	Total Summation	Numeric(15.3)	Same as corresponding value in the 600 record.

** Indicates as specified in the NEM File Format Specification*

Example: 700,ZZZZ000001,214243202,20170302,E1,1,721.308

9) 710 Record

a) A (new) 710 record will be included immediately following each 610 record where the corresponding 200 record contains a UOM = 'kL' and a Suffix = 'W*'.

b) The (new) 710 record will contain the following information separated by a comma (,):-

Table 7	Field	Format	Population Rules
	Record ID	Numeric(3)	'710'
	NMI	Char(10)	NMI* from 200 record
	Meter Serial Number	VarChar(12)	MeterSerialNumber* from 200 record
	Date	Date(8)	IntervalDate* from 300 record
	Channel ID	Char(2)	NMISuffix* from 200 record
	Daily Summation	Numeric(15.3)	'1'
	Total Summation	Numeric(15.3)	Same as corresponding value in the 600 record.

** Indicates as specified in the NEM File Format Specification*

Example: 710,ZZZZ000001,214243202,20170302,W1,1,721.308

10) Filename

a) The file naming convention will comply with the MDFF specification as defined and amended from time-to-time by AEMO.
Example: NEM12#17030300001000000#TCAUSTM#[ToParticipant].csv

b) The 'ToParticipant' will be an agreed 8 character non-market participant code.

11) File Size

a) The file size can be negotiated.

b) Unless agreed to otherwise, the file size will be limited to less than 1MB unzipped in accordance with the B2B procedures.

12) Zipping

a) The eNEM12 files will be zipped prior to delivery.

b) The zipping convention will comply with the MDFF specification as defined and amended from time-to-time by AEMO.

13) File Delivery

a) All files will be delivered via Secure FTP (SFTP) to an agreed address.

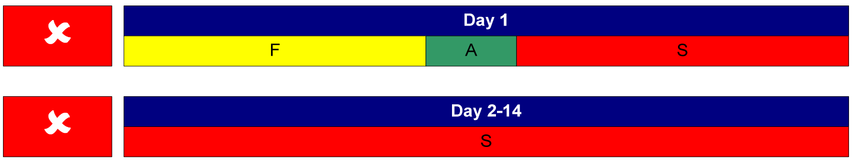
APPROVAL			
Approved By:		**Signature:**	
Position:		**Date:**	
Approved By:		**Signature:**	
Position:		**Date:**	

Scenario 1: New Meter - Commissioning

A new meter is commissioned on day 1 at 10:00am resulting in the following new readings:-

- Zero load final substitutions from 00:00 to 10:00 on day 1;
- Actual data from 10:00 to 11:30 on day 1 as a result of the commissioning read; and
- Forward substitutions from 11:30 on day 1 to 00:00 on day 15 to facilitate communications faults.

Result 1:
No data sent via eNEM12 on account of all days containing non-actual data.



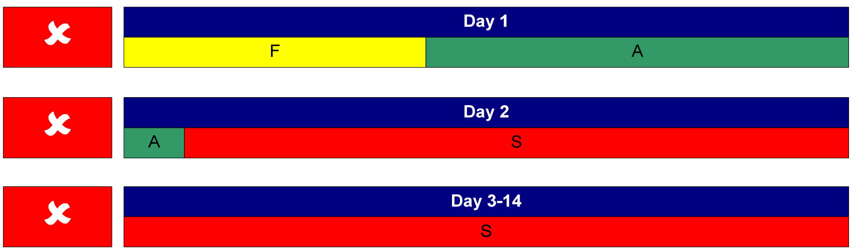
Note 1:
ABS systems create a 'forward' substitution for a period of 14 days following the commissioning of a newly created meter or datastream to facilitate automated substitutions in the event a communications fault occurring before sufficient historic data exists.

Scenario 2: New Meter – Ongoing Reads (Day 2)

The new meter referred to in scenario 1 (above) now reads 02:00 on day 2 resulting in the following new readings:-

- Zero load final substitutions from 00:00 to 10:00 on day 1;
- Actual data from 10:00 on day 1 to 02:00 on day 2; and
- Forward substitutions from 02:00 on day 2 to 00:00 on day 15 to facilitate communications faults.

Result 2:
No data sent via eNEM12 on account of day 1 containing non-actual data.

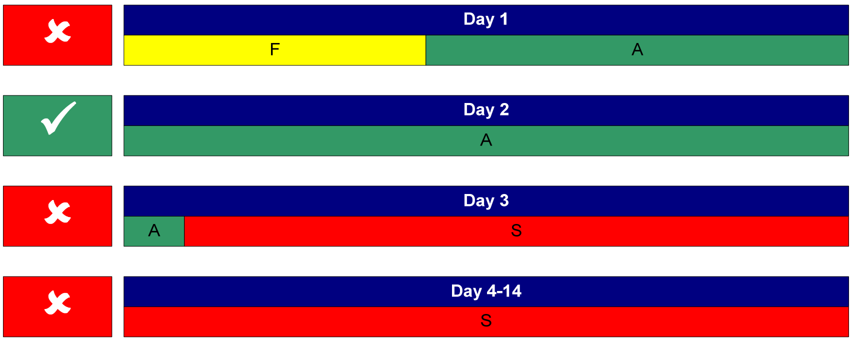


Scenario 3: New Meter – Ongoing Reads (Day 3)

The new meter referred to in scenario 1 and scenario 2 (above) now reads at 03:00 on day 3 resulting in the following new readings:-

- Zero load final substitutions from 00:00 to 10:00 on day 1;
- Actual data from 10:00 on day 1 to 03:00 on day 3; and
- Forward substitutions from 03:00 on day 3 to 00:00 on day 15 to facilitate communications faults.

Result 3:
Data for day 2 is sent via eNEM12. The system identifies day 2 as being the first day to contain all-actual data since the new meter was installed and assumes the 'Total Summation' register is to start from zero. Data for day 1 will never be sent on account of it containing non-actual data.

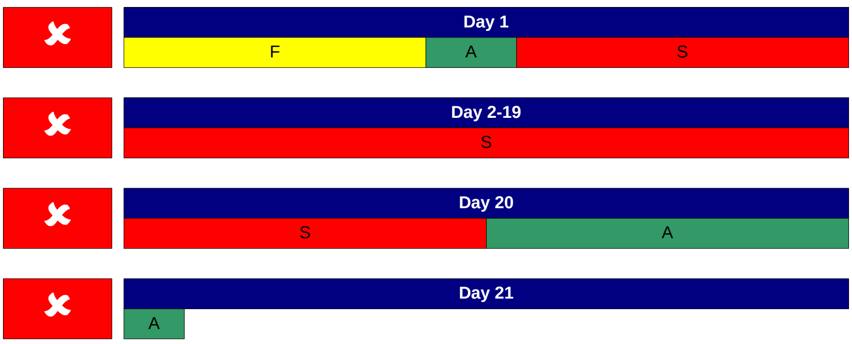


Scenario 4: New Meter – De-Energised After Commissioning

A new meter, similar to that referred to in scenario 1 (above) is commissioned at 10:00 on day 1 and then immediately de-energised to prevent unauthorised use. The NMI is left de-energised until a customer is ready to move in. Whilst de-energised, ABS systems continue to attempt to read the de-energised meter but are forced to substitute on account of a (known) communications fault. The NMI (meter) is energised at 12:00 on day 20 and successfully read at 02:00 on day 21 resulting in the following new readings:-

- Zero load final substitutions from 00:00 to 10:00 on day 1;
- Actual data from 10:00 to 11:30 on day 1 as a result of the commissioning read;
- Temporary substitutions from 11:30 on day 1 to 12:00 on day 20; and
- Actual data from 12:00 on day 20 to 02:00 on day 21;

Result 4:
No data sent via eNEM12 on account of all days containing non-actual data.



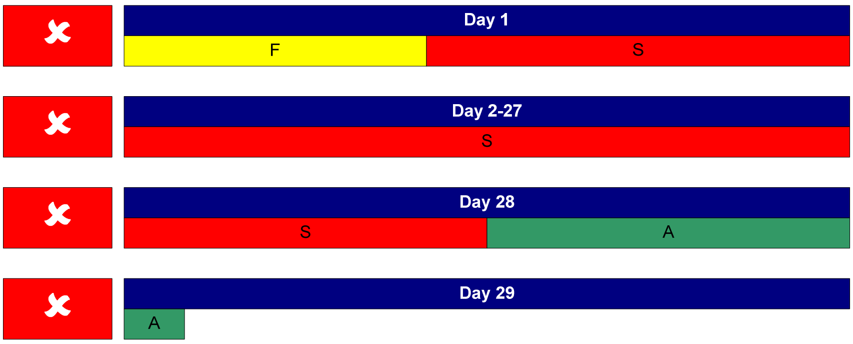
Note 4: Providing the meter reads successfully on day 22, day 21 will be considered the first day to contain all-actual data since the new meter was installed and assumes the 'Total Summation' register is to start from zero.

Scenario 5: New Meter – Greenfield (No Commissioning Read)

A new meter, similar to that referred to in scenario 1 (above) is installed at 10:00 on day 1 on a Greenfield NMI without power. The NMI remains de-energised for a period of time. Whilst de-energised, ABS systems continue to attempt to read the de-energised meter but are forced to substitute on account of a (known) communications fault. The NMI (meter) is energised at 12:00 on day 28 and successfully read at 02:00 on day 29 resulting in the following new readings:-

- Zero load final substitutions from 00:00 to 10:00 on day 1;
- Temporary substitutions from 10:00 on day 1 to 12:00 on day 28; and
- Actual data from 12:00 on day 28 to 02:00 on day 29;

Result 5:
No data sent via eNEM12 on account of all days containing non-actual data



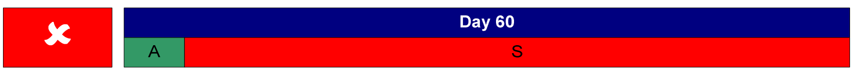
Note 5: Providing the meter reads successfully on day 30, day 29 will be considered the first day to contain all-actual data since the new meter was installed and assumes the 'Total Summation' register is to start from zero.

Scenario 6: Communications Fault - Initial

An established meter reads successfully at 02:00 on day 60 and then fails to at 02:00 on day 61 resulting in the following readings:-

- Actual data from 00:00 on day 60 to 02:00 on day 60; and
- Temporary substitutions from 02:00 on day 60 to 00:00 on day 61.

Result 6:
No data sent via eNEM12 on account of day 60 containing non-actual data.

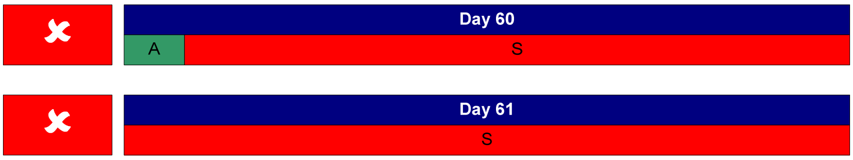


Scenario 7: Communications Fault - Recurring

The established meter that failed to read at 02:00 on day 61 (in scenario 6) fails to read again at 03:00 on day 62 resulting in the following readings:-

- Actual data from 00:00 on day 60 to 02:00 on day 60; and
- Temporary substitutions from 02:00 on day 60 to 00:00 on day 62.

Result 7:
No data sent via eNEM12 on account of day 61 containing non-actual data.

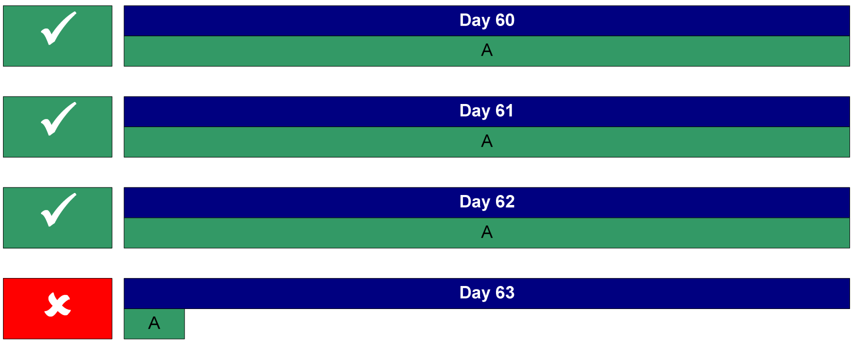


Scenario 8: Communications Fault - Corrected

The established meter that failed to read at 02:00 on day 61 and again at 03:00 on day 62 (in scenarios 6 and 7) now reads successfully at 02:30 on day 63 resulting in the following readings:-

- Actual data from 00:00 on day 60 to 02:30 on day 63.

Result 8:
Data for days 60, 61 and 62 is sent via eNEM12. The system will process each day's data in series so as to sequence the 'Total Summation' register for each day.

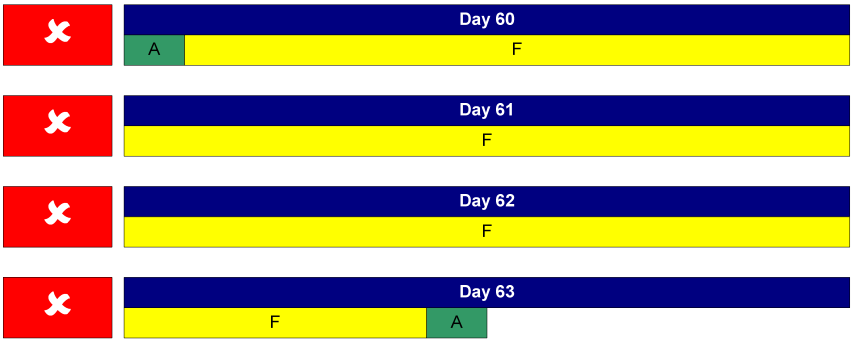


Scenario 9: Communications Fault – Meter Fault

An established meter similar to that referred to in scenario 6 failed to read at 02:00 on day 61 and again at 03:00 on day 62, fails again at 02:30 on day 63 and is consequently identified as being faulty and is replaced at 10:00 on day 63 resulting in the following readings:-

- Actual data from 00:00 on day 60 to 02:00 on day 60;
- Final substitutions from 02:00 on day 60 to 10:00 on day 63; and
- Actual data from 10:00 on day 63 to 11:30 on data 63 as a result of the commissioning read.

Result 9:
No data sent via eNEM12 on account of days 60 to 62 containing non-actual data.



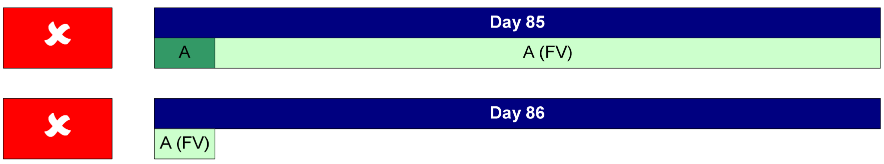
Note 9:
When the new meter reads on day 64, no data will be sent for day 63 on account of it also containing non-actual data. When the new meter reads on day 65, day 64 data will be sent via eNEM12 and this will be interpreted as the first day to contain all-actual data since the new meter was installed and assumes the 'Total Summation' register is to start from zero.

Scenario 10: Failed Validation

An established meter reads successfully at 03:30 on day 85 and again at 02:30 on day 86. The day 86 fails validation resulting in the following readings:-

- Actual data from 00:00 on day 85 to 03:30 on day 85 that has passed validation; and
- Actual data from 03:30 on day 85 to 02:30 on day 86 that has failed validation.

Result 10:
The original version of data for day 85 is not sent via eNEM12 on account of it containing readings that have failed validation.



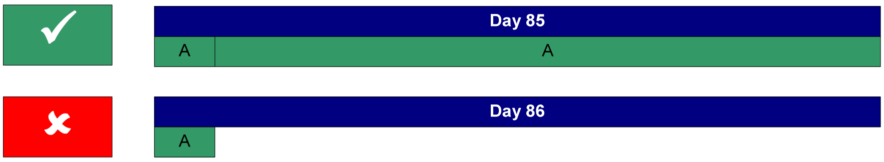
Scenario 11: Failed Validation – Cleared

A user investigates a failed validation exception similar to that referred to in scenario 10 and determines the data is OK and releases the data without any further action resulting in the following readings:-

- Actual data from 00:00 on day 85 to 02:30 on day 86 that has passed validation

Result 11:

The new version of data for day 85 created as a result of the data being released from the validation queue will be sent via eNEM12 in the same way it would have been sent if the data had passed validation.



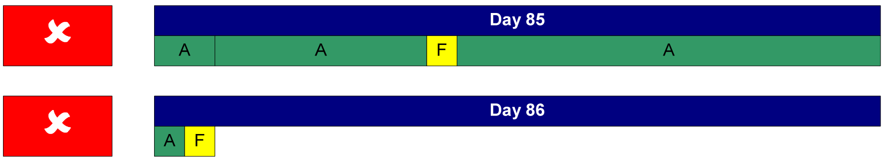
Scenario 12: Failed Validation – Substituted

A user investigates a failed validation exception similar to that referred in scenario 10, and determines all or part of the readings collected in the 02:30 reading on day 86 are erroneous and substitutes the offending intervals resulting in the following readings:-

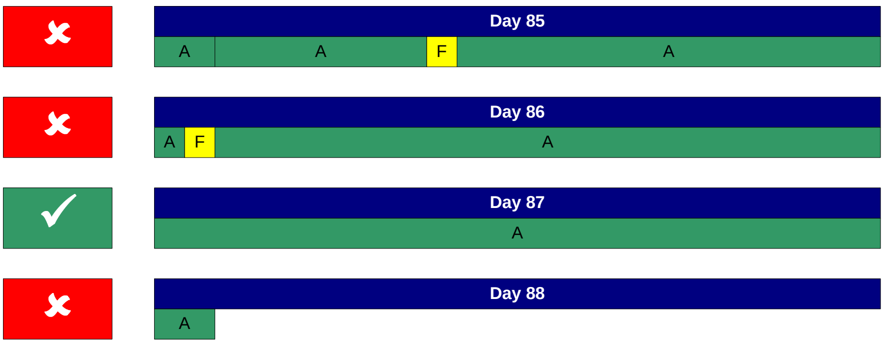
- Actual data from 00:00 on day 85 to 03:30 on day 85; and
- A mixture of actual and/or substituted data from 03:30 on day 85 to 02:30 on day 86.

Result12:

The new version of data created for day 85 as a result of the data being substituted will not be sent via eNEM12 on account of it containing non-actual data.



In addition, should any of the interval data for day 86 require substitution, then day 86 will also be excluded from the eNEM12. Only the next day containing all-actual data will be sent via the eNEM12.



Assuming day 84 contained all-actual data and was sent in the eNEM12, the 'Total Summation' register for day 87 will be the sum of the 'Total Summation' for day 84 plus the consumption for day 87. Not only will days 85 and 86 not be sent via eNEM12, the consumption will not contribute to the calculation of the 'Total Summation' register for subsequent days.

Note 12A:

The validation process is designed to identify corrupt data such as spikes and periods of injected consumption during testing. Standard procedures are designed to capture this and replace it with final substitutions.

Note 12B:

Work instructions will be modified so that any metering data for non-market NMIs (ZZZZ-prefix) that has failed validation will be (simply) manually accepted by the user. This will result in as-read data being provided in the eNEM12 – Albeit outside the normal batch delivery timeframes.

Note 12C:

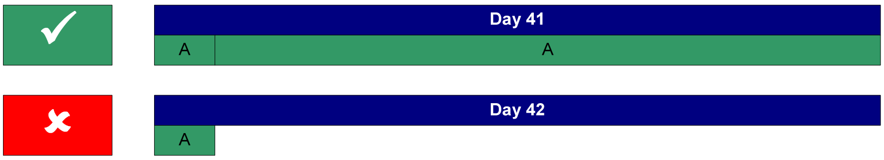
This scenario will only apply to on-market NMIs where the market validation and substitution rules must be applied. Data for any day(s) that has been substituted as a result of the validation process will not be delivered via eNEM12. The 'Total Summation' register will increment on subsequent days as if the total consumption on those days containing substituted data is equal to zero.

Scenario 13: Corrected Data

An established meter reads successfully at 02:00 on day 41 and again at 03:00 on day 42. The day 42 read passes validation resulting in the following readings:-

- Actual data from 00:00 on day 41 to 03:00 on day 42.

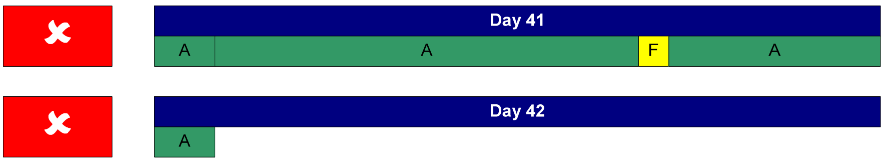
Data for day 41 is sent eNEM12:-



Following the delivery of day 41 data (and possibly the delivery many subsequent days thereafter), an investigation identifies some corrupt data interval(s) on day 41 resulting final substitutions replacing actual data.

Result13:

The new version of data created for day 41 as a result of the data being final substituted will not be sent via eNEM12 on account of it containing non-actual data.



Note 13A:
From time-to-time queries relating to billing and/or metering data result in actual data being over-written with final substitutions. The suppression of the delivery of non-actual data will result in the new version of data not being delivered and all calculations of the 'Total Summation' register continuing to include the erroneous data.

Note 13B:

The retailer is committed to resolving billing disputes for non-market NMIs (ZZZZ-prefix) internally within either the billing system or billing process. Whilst the MDP may be asked to investigate spurious data, there is no requirement to edit and re-send any data as a result of those investigations.

Note 13C:

This scenario will only apply to on-market NMIs where the market validation and substitution rules must be applied. Data for any day(s) that has been substituted after being delivered as actual data on on-market NMIs will not be delivered via eNEM12. The system will continue to accumulate the 'Total Summation' register as if the edit never occurred.